

## Research paper

# Low-carbon economic dispatch of integrated energy system with CCS-P2G-CHP

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## ABSTRACT

Integrated energy system (IES) is an important way and main starting point to achieve the goal of double carbon. In view of this, this paper constructs a low-carbon economic dispatch model for IES, including the carbon capture and storage (CCS), the two-stage power-to-gas (P2G) and the combined heat and power (CHP) unit. Firstly, at the technical level, the mathematical models of CCS, two-stage P2G and CHP are constructed for the CCS-P2G-CHP coupling system. Secondly, at the market mechanism level, the ladder carbon trading model is employed to constraint carbon emissions of IES. Finally, an optimal scheduling strategy with the objective of minimizing the total operation cost of IES is proposed, which is solved by the IPOPT solver in MATLAB. The effectiveness of the proposed model is verified by setting up multiple scenarios for example analysis, and the influence of the sensitivity of each step carbon trading parameter of the low carbon and economy of IES is analyzed.

## 1. Introduction

With the extensive utilization of fossil energy, the world is facing an increasingly serious energy crises and environmental pollutions (Chatopadhyay et al., 2021), climate change caused by greenhouse gas emissions (CO<sub>2</sub> et al.) is the biggest threat to human survival. The large-scale development and utilization of renewable energy sources and low-carbon technologies provide effective pathways to solve the abovementioned problems, but the uncertainty intensifies of renewable energy and the immaturity of low-carbon technologies lead to energy waste and high dispatch costs (Hou et al., 2021; Wang et al., 2022). Integrated energy system (IES) integrated multiple energy networks, which is an effective method for achieving economic performance and low-carbon development of energy systems (Alabdulwahab et al., 2015), attracting wide attention from researchers at home and abroad (Song et al., 2022; Lonis et al., 2021).

In order to realize the decarbonization of energy systems and meet the needs of users for electric energy, combined heat and power (CHP) are widely used (Maghanki et al., 2013; Bagherian and Mehranzamir, 2020). Reference (Guo et al., 1996) decomposed CHP economic dispatch

problem into the heat dispatch and the power dispatch, which were connected by the heat-power feasible region constraints of co-generation units, and used a two-layer algorithm to solve the dispatch problem. The authors in (Vasebi et al., 2007) presented a harmony search algorithm to solve the combined heat and power economic dispatch problem, and analyzed the strength compared to conventional methods. The characteristics of biogas in CHP generation were analyzed (Hosseini and Wahid, 2014). In (Zou and Gong, 2022), a differential evolution using migrated variables was presented to tackle with CHP dynamic economic dispatch problems. The above studies have focused on the economy of the system, the low carbon attribute has not been studied in depth.

Power-to-gas (P2G) technology can utilize electricity to produce hydrogen or synthetic natural gas (Clegg and Mancarella, 2015), which is widely introduced into IES to promote the integration and accommodation of renewable energy sources (Yang et al., 2018). An IES optimal scheduling model with P2G under uncertainties was constructed in (Zhou et al., 2020), and developed a two-stage robust optimization method to deal with the system operation risk. Reference (Zhang et al., 2022a) considered the different functional characteristics of P2G

**Abbreviations:** IES, Integrated energy system; CCS, Carbon capture and storage; P2G, Power-to-gas; CHP, Combined heat and power; SNG, Synthetic natural gas; MT, Micro-turbines; EL, Electrolyzer; MR, Methane Reactor; HFC, Hydrogen fuel cell; ER, Electric refrigerator; CCPP, Carbon-capture power plants; PDR, Price-Based demand response.

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technology, established a highly decoupled multi-step optimization model of residential IES, and adopted variable method to linearize the model. In (Zhao et al., 2020), P2G facilities were used for voltage management by mitigating fluctuating photovoltaic (PV) power output via converting the surplus generation into hydrogen, which prompts the accommodation of PV power. A model with the coupled operation of CCPP-P2G and the PDR considering dynamic carbon trading mechanisms was established in (Cui et al., 2023), which can flexibly capture the supply and demand fluctuations in the carbon trading market. Reference (Wang et al., 2021) divided traditional P2G technology into two-stage P2G technology and SNG technology) based on the inherent advantages of hydrogen energy, which makes full use of the advantages of hydrogen and improves the efficiency of P2G.

Carbon capture and storage (CCS) has been regarded as an effective method to reduce carbon dioxide emissions from power generation facilities (Kang et al., 2011; Ma et al., 2011). Reference (Dalla Longa et al., 2020) assessed how reductions of the costs of CCS processes affect the deployment of it in the various sectors of the economy. In (Zhang et al., 2022b), a model of carbon-cycle integrated energy system with CCS, P2G and S-CO<sub>2</sub> was constructed, and analyzed CCS can increase the direct operating and pollutants treatment costs as well as the penetration rate of renewable energy. In (Chen et al. 2023), a CHP-CCS-P2G low carbon economic dispatch model was constructed by combining source-side scheduling and demand-side scheduling. By reasonably scheduling flexible loads on the demand side, the system's peak shaving pressure was reduced. An optimal scheduling model for CHP with P2G and CCS was established in (Ma et al., 2021; Zhang et al., 2022), which solved the problems of the carbon source required for P2G and the CHP's carbon emissions by the optimal dispatch in IES. However, in this dispatch model, hydrogen does not be considered which caused the efficiency of P2G facilities is low. The comparison of the proposed model with related works are shown in Table 1.

In order to achieve the low-carbon and economic operation of IES, this paper introduces the corresponding carbon reduction measures from two aspects of technology and market. At the technology level, CCS, P2G and CHP technology are introduced into IES; at the market level, carbon trading mechanism is designed to constraint the emissions of carbon dioxide. The main contributions of this paper are summarized as follows:

- 1) The model of CCS, two-stage P2G and CHP, and their coupling mechanism are established.

**Table 1**  
Comparison of the proposed model with related works.

| Ref                                      | Method                                          | CCS | P2G | CHP | Carbon trading |
|------------------------------------------|-------------------------------------------------|-----|-----|-----|----------------|
| (Guo et al., 1996)                       | Two-Layer algorithm                             | ×   | ×   | ✓   | ×              |
| (Vasebi et al., 2007)                    | harmony search algorithm                        | ×   | ×   | ✓   | ×              |
| (Zou and Gong, 2022)                     | differential evolution using migrated variables | ×   | ×   | ✓   | ×              |
| (Zhou et al., 2020)                      | Two-Stage robust optimization                   | ×   | ✓   | ×   | ✓              |
| (Zhang et al., 2022a)                    | variable method                                 | ×   | ✓   | ×   | ✓              |
| (Cui et al., 2023; Wang et al., 2021)    | Mixed linear integer programming                | ✓   | ✓   | ×   | ✓              |
| (Ma et al., 2011)                        | Mixed linear integer programming                | ✓   | ×   | ×   | ✓              |
| (Dalla Longa et al., 2020)               | Mixed linear integer programming                | ✓   | ✓   | ×   | ×              |
| (Zhang et al., 2022b; Chen et al., 2023) | Mixed linear integer programming                | ✓   | ✓   | ✓   | ×              |
| This work                                | Mixed linear integer programming                | ✓   | ✓   | ✓   | ✓              |

- 2) A low-carbon economic dispatch model of IES with CCS-P2G-CHP considering the ladder carbon trading mechanism is constructed.
- 3) The operation cost, the rate of renewable energy accommodation as well as the carbon emissions of IES are analyzed.

The remainder of this article is summarized as follows: Section II analyzes the coupling mechanism of CHP-P2G-CCS. Section III establishes the model of carbon trading mechanism. Section IV proposes a low-carbon economic dispatch model of IES with CHP-P2G-CCS coupling. Section V presents the simulation results and corresponding analysis. Section VI concludes this article.

## 2. IES structure with CCS-P2G-CHP coupling

The IES with CCS-P2G-CHP coupling constructed in this paper consists of wind turbine (WT), photovoltaic (PV) unit, CCS, two-stage P2G, CHP unit, MT, electric refrigerator (ER), natural gas network and cooling, heating, power and gas load. The specific frame diagram is shown in Fig. 1.

CCS captures CO<sub>2</sub> produced by CHP and MT and the carbon source required for the two-stage P2G methanation reaction. In the two-stage P2G, the electric energy is converted into hydrogen energy through the EL. Part of the hydrogen energy is input into the MR for methanation reaction with the CO<sub>2</sub> captured by CCS. The natural gas synthesized by the methanation reaction is supplied to the gas load and MT, and the other part of the hydrogen energy is directly converted into electric energy and heat energy through HFC.

### 2.1. Modeling of CCS

The electrical power consumed by CCS is labeled as  $P_{CCS, e}$ , which contains the basic energy consumption  $P_{B, e}$  and the operation energy consumption  $P_{R, e}$ . The basic energy consumption is independent of the operating condition of the carbon capture system and can be regarded as a constant. The specific expression is as follows:

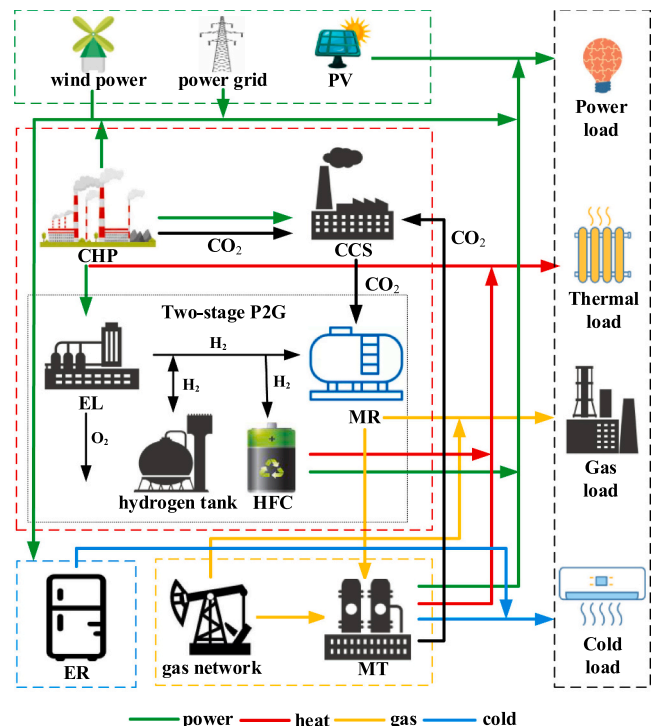


Fig. 1. IES frame diagram.

$$\begin{cases} P_{CCS,t} = P_{R,t} + P_{B,t} \\ P_{R,t} = \sigma E_{ZST,t} \end{cases} \quad (1)$$

where  $\sigma$  is the energy consumption required to capture per unit of CO<sub>2</sub> and  $E_{ZST,t}$  is the amount of CO<sub>2</sub> treated by the regeneration tower in CCS at time  $t$ :

$$\begin{cases} E_{CCS,t} = E_{JZ,t} - E_{air} \\ E_{JZ,t} = \chi (P_{CHP,t} + P_{MT,t}) \end{cases} \quad (2)$$

where  $E_{CCS,t}$  is the amount of CO<sub>2</sub> absorbed by CCS at time  $t$ ,  $E_{JZ,t}$  is the emission of the system unit at time  $t$ ,  $E_{air}$  is the amount of CO<sub>2</sub> discharged into the atmosphere at time  $t$ ,  $\chi$  is the calculation coefficient of CO<sub>2</sub> (He et al., 2018),  $P_{CHP,t}$  is the electric power emitted by CHP at time  $t$ ,  $P_{MT,t}$  is the electric energy output by MT at time  $t$ .

The electrical power constraint of CCS is:

$$P_{CCS, \min} \leq P_{CCS,t} \leq P_{CCS, \max} \quad (3)$$

where  $P_{CCS, \min}$  and  $P_{CCS, \max}$  are the lower and upper limits of CCS electric power, respectively.

## 2.2. Modeling of two-stage P2G

In order to accurately reflect the operating condition of P2G, this paper replaces the traditional P2G model with EL, MR, and HFC, and refines it into two processes: electric hydrogen production and methanation. EL converts electrical energy into hydrogen energy. Part of the hydrogen energy is input into MR and CO<sub>2</sub> captured by CCS for methanation reaction, and the other part of the hydrogen energy is directly converted into electrical energy and heat energy through HFC. The specific model is as follows.

The relationship between the hydrogen energy produced by EL and the electricity it consumes is:

$$H_{EL,t} = \eta_{H_2} P_{P2G,t} \quad (4)$$

where  $H_{EL,t}$  is the hydrogen energy output by EL at time  $t$ ,  $\eta_{H_2}$  is the hydrogen energy conversion efficiency,  $P_{P2G,t}$  is the electrical power consumed by P2G at time  $t$ .

The P2G power constraint is:

$$P_{P2G, \min} \leq P_{P2G,t} \leq P_{P2G, \max} \quad (5)$$

where  $P_{P2G, \min}$  and  $P_{P2G, \max}$  are the lower and upper limits of P2G electric power respectively.

The gas power  $Q_{MR,t}$  converted by MR is expressed as:

$$Q_{MR,t} = \eta_{MR} H_{MR,t} \quad (6)$$

where  $\eta_{MR}$  is the MR conversion efficiency,  $H_{MR,t}$  is the hydrogen power consumed by MR at time  $t$ .

The amount of CO<sub>2</sub> required in the methanation process  $EMR,t$  is expressed as:

$$EMR,t = \sum_{t=1}^T \chi Q_{MR,t} \quad (7)$$

The CO<sub>2</sub> required in the methanation reaction is provided by CCS:

$$E_{CCS,t} = EMR,t \quad (8)$$

The electrical power and thermal power of HFC output are expressed as:

$$P_{HFC,t} = \eta_{HFC}^e H_{HFC,t} \quad (9)$$

$$R_{HFC,t} = \eta_{HFC}^r H_{HFC,t} \quad (10)$$

where  $P_{HFC,t}$  and  $R_{HFC,t}$  are the electrical power and thermal power generated by HFC at time  $t$ ,  $\eta_{HFC}^e$  and  $\eta_{HFC}^r$  are the electric and thermal

conversion efficiency of HFC, respectively, and  $H_{HFC,t}$  is the hydrogen power input to HFC at time  $t$ .

## 2.3. Modeling of CCS-P2G-CHP coupling system

Based on the above research, this paper combines CCS, two-stage P2G and CHP as a whole to establish a CCS-P2G-CHP coupling model. Part of the CO<sub>2</sub> produced during the operation of CHP and MT is discharged into the atmosphere through the flue gas, and the other part is captured by CCS. The captured CO<sub>2</sub> is used as a carbon source for the two-stage P2G methanation reaction. The operating characteristic of CCS-P2G-CHP coupling system is shown in Fig. 2. Fig. 2(a) depicts the coupling relation between CHP, CCS and P2G. Fig. 2(b) depicts the comparison of output power of the CHP model and the CCS-P2G-CHP coupling model. It can be seen that the coupling of CCS-P2G-CHP can facilitate the utilization of CO<sub>2</sub> while decrease the electro-thermal coupling characteristics of CHP as well as enhance the power regulation capability of CHP.

CCS and P2G power are supplied by CHP:

$$P_{CHP,t} = P_{P2G,t} + P_{CCS,t} + P_{E,t} \quad (11)$$

where  $P_{E,t}$  represents the power supplied by the grid.

The thermoelectric characteristic of CHP is expressed as:

$$\begin{aligned} \max \{ & P_{CHP, \min} - Z_{v1} R_{CHP,t}, S_m (R_{CHP,t} - R_{CHP0}) \} \\ \leq & P_{P2G,t} + P_{CCS,t} + P_{E,t} \leq P_{CHP, \max} - Z_{v2} R_{CHP,t} \end{aligned} \quad (12)$$

where  $Z_{v1}$  and  $Z_{v2}$  are the thermoelectric conversion coefficients corresponding to the minimum and maximum output power, respectively.  $R_{CHP,t}$  is the thermal power of CHP at time  $t$ ,  $R_{CHP0}$  is the thermal power with the minimum electrical power, and  $S_m$  is the linear supply slope of CHP electrical power and thermal power.

The electrical power constraint of CHP is:

$$P_{CHP, \min} \leq P_{CHP,t} \leq P_{CHP, \max} \quad (13)$$

where  $P_{CHP, \min}$  and  $P_{CHP, \max}$  are the lower and upper limits of CHP electrical power, respectively.

The thermal power constraint of CHP is:

$$R_{CHP, \min} \leq R_{CHP,t} \leq R_{CHP, \max} \quad (14)$$

where  $R_{CHP, \min}$  and  $R_{CHP, \max}$  are the lower and upper limits of CHP thermal power respectively.

The power range provided by CHP with CCS and P2G to the grid is

$$\begin{aligned} P_{E, \min} = & P_{CHP, \min} - P_{P2G, \max} - P_{CCS, \max} \leq P_{E,t} \\ \leq & P_{CHP, \max} - P_{P2G, \min} - P_{CCS, \min} = P_{E, \max} \end{aligned} \quad (15)$$

where  $P_{E, \min}$  and  $P_{E, \max}$  are the lower and upper limits of the grid supply power in the CCS-P2G-CHP coupling system respectively.

The thermoelectric coupling characteristics of CCS-P2G-CHP can be expressed as:

$$\begin{aligned} \max \left\{ & P_{CHP, \min} - Z_{v1} R_{CHP,t} - P_{P2G,t} - P_{CCS,t} \right\} \\ & S_m (R_{CHP,t} - R_{CHP0}) - P_{P2G,t} - P_{CCS,t} \\ \leq & P_{E,t} \leq P_{CHP, \max} - Z_{v2} R_{CHP,t} - P_{P2G,t} - P_{CCS,t} \end{aligned} \quad (16)$$

Substituting Eq. (3) and Eq. (5) into Eq. (16), the thermoelectric coupling characteristics of CCS-P2G-CHP can be further rewritten as:

$$\begin{aligned} \max \left\{ & P_{E, \min} - Z_{v1} R_{CHP,t}, S_m (R_{CHP,t} - R_{CHP0}) \right\} \\ & - P_{P2G, \max} - P_{CCS, \max} \\ \leq & P_{E,t} \leq P_{E, \max} - Z_{v2} R_{CHP,t} - P_{P2G, \min} - P_{CCS, \max} \end{aligned} \quad (17)$$

According to Eqs. (14), (15) and (17), by introducing CCS and P2G into CHP, the minimum electric power of CHP unit is reduced from  $P_{CHP, \min}$  to  $P_{E, \min}$ , the output range of electric power is increased, the

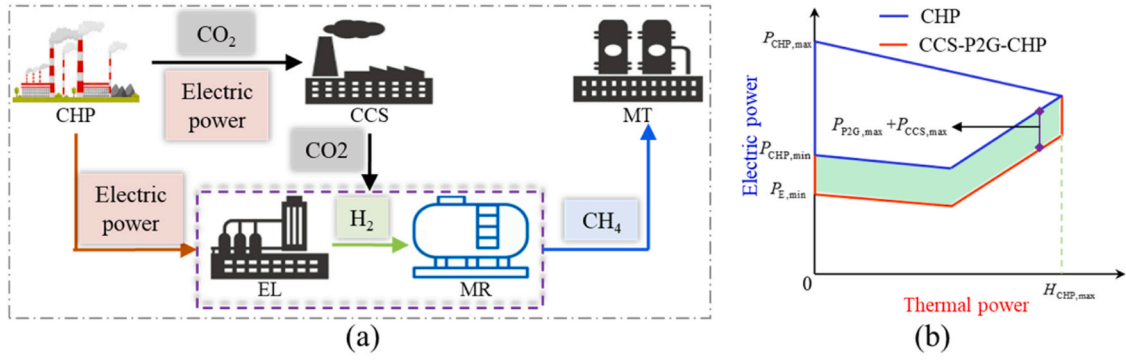


Fig. 2. Operating characteristic of CCS-P2G-CHP coupling system.

adjustable ability is enhanced, and the coupling characteristics are reduced.

Substituting Eq. (4) into Eq. (5), the hydrogen energy range in the CCS-P2G-CHP model is expressed as:

$$\begin{aligned} H_{EL,min} &= \eta_{H_2} P_{P2G,min} \leq H_{EL,t} \\ &\leq \eta_{H_2} P_{P2G,max} = H_{EL,max} \end{aligned} \quad (18)$$

where  $H_{EL,min}$  and  $H_{EL,max}$  are the lower and upper limits of hydrogen energy in CCS-P2G-CHP model respectively.

From Eq. (16) and Eq. (17), it can be seen that in the CCS-P2G-CHP model, the hydrogen energy is limited by the electrical power provided by CHP to the grid. When electrical power of the grid takes the minimum value, the hydrogen energy produced by EL reaches the maximum value.

Regardless of the effect of CCS basic energy consumption on system operation, let  $P_{CCS,t} = P_{R,b}$ , then the relationship between electricity, heat, and hydrogen in the CCS-P2G-CHP model is expressed as:

$$\begin{aligned} &\left\{ \frac{\eta_{H_2}}{1 + \sigma_{\chi}} \left[ (P_{CHP,min} - Z_{v1} R_{CHP,t} - P_{E,t} - D) \right], \right. \\ &\left. S_m (R_{CHP,t} - R_{CHP0}) - P_{E,t} - D \right\} \\ &\leq H_{EL,t} \leq \frac{\eta_{H_2}}{1 + \sigma_{\chi}} \left( P_{CHP,max} - \right. \\ &\left. Z_{v2} R_{CHP,t} - P_{E,t} - D \right) \end{aligned} \quad (19)$$

$$D = \sigma_{\chi} (P_{E,t} + P_{CCS,t} + P_{MT,t})$$

It can be seen from t that the adjustable range of CHP containing CCS and P2G increases and the coupling characteristic decreases.

### 3. Ladder carbon trading mechanism

The carbon trading mechanism is the abbreviation of the carbon emission trading mechanism, that is, the government or regulatory authorities formulate carbon emission rights according to the emission reduction targets and allocate them to each enterprise included in the carbon trading system. Each enterprise can put the valuable assets of carbon emission rights as commodities in the secondary market for open and free trading.

#### 3.1. Actual carbon emissions

The actual emissions of CO2 should remove the part captured by CCS:

$$\begin{aligned} E_{CO_2,t} &= \varepsilon_1 (P_{CHP,t} + Z_{v1} R_{CHP,t}) \\ &+ \varepsilon_2 (P_{CHP,t} + Z_{v1} R_{CHP,t})^2 \\ &+ \varepsilon_3 + \varepsilon_4 P_{MT,t} - E_{CCS,t} \end{aligned} \quad (20)$$

where  $E_{CO_2,t}$  is the actual carbon emission of IES at time t,  $\varepsilon_1$ ,  $\varepsilon_2$ ,  $\varepsilon_3$  are the carbon dioxide emission coefficients of CHP, and  $\varepsilon_4$  is the carbon dioxide emission coefficient of MT (Li et al., 2020).

#### 3.2. Carbon emissions allowance

The objects with carbon emission allowance are CHP, MT WT and PV unit.

$$E_{0,t} = Y (P_{CHP,t} + P_{MT,t} + P_{wind,t} + P_{pv,t}) \quad (21)$$

where  $E_{0,t}$  is the initial carbon allowance of IES at time t, Y is the unit power carbon emission allowance, and the weighted average of the regional power marginal emission coefficient and the capacity marginal coefficient is 0.798.  $P_{wind,t}$  is the output of the WT at time t, and  $P_{pv,t}$  is the output of the photovoltaic unit at time t.

#### 3.3. Ladder carbon transaction cost model

Carbon emissions involved in the carbon trading mechanism is:

$$E_{IES,t} = E_{CO_2,t} - E_{0,t} \quad (22)$$

Compared with the traditional carbon trading mechanism, the ladder carbon trading mechanism adopts a ladder pricing mechanism to divide the purchase interval. As the demand for carbon emission allowance increases, the purchase price gradually increases to constrain carbon emissions. When the carbon emission quota is higher than the carbon emission, the surplus quota can be sold for profit. In order to improve the enthusiasm of emission reduction, the compensation coefficient is introduced to increase the degree of reward and stimulate emission reduction.

The ladder carbon transaction cost calculation model is:

$$f_{CO_2,t} = \begin{cases} \xi E_{IES,t}, & E_{IES,t} \leq L \\ \xi(1 + \tau)(E_{IES,t} - L) + \xi L, & L < E_{IES,t} \leq 2L \\ \xi(1 + 2\tau)(E_{IES,t} - 2L) + \xi(2 + \tau)L, & 2L < E_{IES,t} \leq 3L \\ \xi(1 + 3\tau)(E_{IES,t} - 3L) + \xi(3 + 3\tau)L, & 3L < E_{IES,t} \leq 4L \\ \xi(1 + 4\tau)(E_{IES,t} - 4L) + \xi(4 + 6\tau)L, & E_{IES,t} \geq 4L \end{cases} \quad (23)$$

Where  $f_{CO_2,t}$  is the carbon transaction cost at time t,  $\xi$  is the base price of carbon trading, L is the length of carbon emission interval,  $\tau$  is the price growth rate.

### 4. IES low-carbon economic dispatch model with CCS-P2G-CHP coupling

#### 4.1. Objective function

The optimization goal of IES with CCS-P2G-CHP coupling is to minimize the total operating cost, including the carbon transaction cost, operation cost of CCS-P2G-CHP coupling model, MT fuel cost, WT power curtailment penalty cost, PV power curtailment penalty cost, electric

refrigerator operation cost and carbon utilization benefit, as shown in the following:

$$\min F = F_{\text{CO}_2} + F_{\text{CHP}} + F_{\text{MT}} + F_{\text{cwind}} + F_{\text{cpv}} + F_{\text{ER}} \quad (24)$$

where  $F$  is the total operating cost of the system,  $F_{\text{CO}_2}$  is the carbon transaction cost,  $F_{\text{CHP}}$  is the operating cost of the CCS-P2G-CHP coupling system,  $F_{\text{MT}}$  is the fuel cost of MT,  $F_{\text{cwind}}$  is the WT power curtailment penalty cost,  $F_{\text{cpv}}$  is the PV power curtailment penalty cost, and  $F_{\text{ER}}$  is the operating cost of the electric refrigerator.

#### 1) Carbon transaction costs

$$F_{\text{CO}_2} = \sum_{t=1}^T f_{\text{CO}_2} \quad (25)$$

#### 2) CCS-P2G-CHP operation cost P2G operating cost:

$$F_{\text{P2G}} = \sum_{t=1}^T m_1 P_{\text{P2G},t} + r_1 C_{\text{MR},t} \quad (26)$$

where  $m_1$  is the operation and maintenance cost coefficient of P2G, and  $r_1$  is the cost coefficient of  $\text{CO}_2$  required for methanation reaction in P2G (Saboori and Hemmati, 2016).

#### CCS operating cost:

$$F_{\text{CCS}} = \sum_{t=1}^T m_2 P_{\text{CCS},t} \quad (27)$$

where  $m_2$  is the operation and maintenance cost coefficient of CCS.

#### The operating cost CHP is:

$$F_{\text{CHP}} = \sum_{t=1}^T \left[ \frac{a(P_{\text{CHP},t} + Z_{v1} R_{\text{CHP},t})}{+b(P_{\text{CHP},t} + Z_{v1} R_{\text{CHP},t})^2} \right] + m_1 P_{\text{P2G},t} + m_2 P_{\text{CCS},t} + c \quad (28)$$

where  $a$ ,  $b$  and  $c$  are the operating cost coefficients of the CCS-P2G-CHP coupled system (Li et al., 2019).

#### 3) MT fuel cost

The model of MT is:

$$\begin{cases} R_{\text{MT},t} = \eta_{\text{hl}} \eta_{\text{h}} \frac{P_{\text{MT},t}(1 - \eta_{\text{MT}} - \eta_{\text{w}})}{\eta_{\text{MT}}} \\ P_{\text{MT},t} = \eta_{\text{MT}} Q_{\text{MT},t} \\ C_{\text{MT},t} = \eta_{\text{hl}} \eta_{\text{c}} \frac{P_{\text{MT},t}(1 - \eta_{\text{MT}} - \eta_{\text{w}})}{\eta_{\text{MT}}} \end{cases} \quad (29)$$

$$F_{\text{MT}} = \sum_{t=1}^T c_1 Q_{\text{MT},t} \quad (30)$$

where  $Q_{\text{MT},t}$  is the gas power consumed by MT at time  $t$ ,  $\eta_{\text{MT}}$  is the conversion efficiency of MT,  $R_{\text{MT},t}$  is the heating power of the bromine cooler,  $\eta_{\text{hl}}$  is the heating loss coefficient,  $\eta_{\text{h}}$  is the heating coefficient of the bromine cooler,  $\eta_{\text{w}}$  is the flue gas waste heat recovery rate of the bromine cooler,  $C_{\text{MT},t}$  is the cooling power of the bromine cooler,  $\eta_{\text{c}}$  is the refrigeration coefficient of the bromine cooler,  $c_1$  is the cost coefficient of MT (Li et al., 2019).

#### 4) WT power curtailment penalty cost

$$F_{\text{cwind}} = \sum_{t=1}^T c_2 P_{\text{cwind},t} \quad (31)$$

where  $c_2$  is the penalty cost coefficient of WT power,  $P_{\text{cwind},t}$  is the abandoned WT power at time  $t$ .

#### 5) PV power curtailment penalty cost

$$F_{\text{cpv}} = \sum_{t=1}^T c_3 P_{\text{cpv},t} \quad (32)$$

where  $c_3$  is the penalty cost coefficient of PV power abandonment,  $P_{\text{cpv},t}$  is the abandoned PV power at time  $t$ .

#### 6) Operation cost of ER

$$F_{\text{ER}} = \sum_{t=1}^T c_4 P_{\text{er},t} \quad (33)$$

where  $c_4$  is the operation and maintenance cost coefficient of ER, and  $P_{\text{er},t}$  is the electrical power of ER at time  $t$ .

### 4.2. Constraints

#### 1) Electrical power balance

$$P_{\text{HFC},t} + P_{\text{wind},t} + P_{\text{pv},t} + P_{\text{E},t} + P_{\text{MT},t} + P_{\text{CHP},t} = P_{\text{L},t} + P_{\text{er},t} \quad (34)$$

where  $P_{\text{L},t}$  is the electrical load at time  $t$ .

#### 2) Thermal power constraint

$$a_{\text{r},\min} R_{\text{L},t} \leq R_{\text{CHP},t} + R_{\text{HFC},t} + R_{\text{MT},t} \leq a_{\text{r},\max} R_{\text{L},t} \quad (35)$$

where  $a_{\text{r},\max}$ ,  $a_{\text{r},\min}$  are the upper and lower limits of the adjustment ratio in the cold network (Ma et al., 2021),  $R_{\text{L},t}$  is the heat load at time  $t$ .

#### 3) Cold power constraint

$$a_{\text{c},\min} C_{\text{L},t} \leq C_{\text{er},t} + C_{\text{MT},t} \leq a_{\text{c},\max} C_{\text{L},t} \quad (36)$$

where  $a_{\text{c},\max}$ ,  $a_{\text{c},\min}$  are the upper and lower limits of the adjustment ratio in the cold network (Ma et al., 2021),  $C_{\text{L},t}$  is the cooling load at time  $t$ .

#### 4) Gas power constraint

$$Q_{\text{L},t} + Q_{\text{MT},t} \leq Q_{\text{MR},t} + Q_{\text{s},t} \quad (37)$$

where  $Q_{\text{L},t}$  is the gas load at time  $t$ ,  $Q_{\text{s},t}$  is the gas power output by the natural gas source at time  $t$ .

#### 5) Hydrogen power balance

$$H_{\text{EL},t} = H_{\text{MR},t} + H_{\text{HFC},t} + H_{\text{s},t} \quad (38)$$

where  $H_{\text{s},t}$  is the hydrogen energy input to the energy storage device.

#### 6) CCS-P2G-CHP power constraint

$$\lambda_{\text{l}} \leq (P_{\text{P2G},t+1} + P_{\text{CCS},t+1} + P_{\text{E},t+1}) - (P_{\text{P2G},t} + P_{\text{CCS},t} + P_{\text{E},t}) \leq \lambda_{\text{u}} \quad (39)$$

where  $\lambda_{\text{u}}$  and  $\lambda_{\text{l}}$  are the climbing upper and lower limits of the CCS-P2G-CHP coupling system respectively (Li et al., 2019).

#### 7) $\text{CO}_2$ capture constraint of CCS

$$\begin{aligned} E_{\text{CCS},t} &\leq \varepsilon_1 \left( \frac{P_{\text{P2G},t} + P_{\text{CCS},t} + P_{\text{E},t}}{+Z_{v1} R_{\text{CHP},t}} \right) + \\ &\varepsilon_2 (P_{\text{P2G},t} + P_{\text{CCS},t} + P_{\text{E},t} + Z_{v1} R_{\text{CHP},t})^2 \\ &+ \varepsilon_3 + \varepsilon_4 P_{\text{MT},t} \end{aligned} \quad (40)$$

#### 8) Operation constraints of micro gas turbine

$$\begin{cases} P_{\text{MT},\min} \leq P_{\text{MT},t} \leq P_{\text{MT},\max} \\ \lambda_{\text{l},\text{MT}} \leq P_{\text{MT},t+1} - P_{\text{MT},t} \leq \lambda_{\text{u},\text{MT}} \end{cases} \quad (41)$$

where  $P_{\text{MT},\max}$ ,  $P_{\text{MT},\min}$  are the upper and lower limits of the electrical power of MT,  $\lambda_{\text{u},\text{MT}}$ ,  $\lambda_{\text{l},\text{MT}}$  are the upper and lower limits of the climbing of MT (He et al., 2018).

#### 9) Operation constraint of ER

$$P_{\text{er},\min} \leq P_{\text{er},t} \leq P_{\text{er},\max} \quad (42)$$



where  $P_{er, \max}$ ,  $P_{er, \min}$  are the upper and lower limits of the power of ER.

#### 10) Natural gas source constraint

$$Q_{s, \min} \leq Q_{s, t} \leq Q_{s, \max} \quad (43)$$

where  $Q_{s, \max}$ ,  $Q_{s, \min}$  are the upper and lower power limits of the natural gas source respectively.

#### 11) Electrolytic cell operation constraint

$$\lambda_{l, EL} \leq P_{P2G, t+1} - P_{P2G, t} \leq \lambda_{u, EL} \quad (44)$$

where  $\lambda_{u, EL}$ ,  $\lambda_{l, EL}$  are the upper and lower limits of EL respectively.

#### 12) Methane reactor operating constraint

$$\begin{cases} H_{MR, \min} \leq H_{MR, t} \leq H_{MR, \max} \\ \lambda_{l, MR} \leq H_{MR, t+1} - H_{MR, t} \leq \lambda_{u, MR} \end{cases} \quad (45)$$

where  $H_{MR, \max}$ ,  $H_{MR, \min}$  are the upper and lower limits of the hydrogen power input into MR,  $\lambda_{u, MR}$ ,  $\lambda_{l, MR}$  are the upper and lower limits of MR climbing, respectively.

#### 13) Hydrogen fuel cell operation constraints

$$\begin{cases} H_{HFC, \min} \leq H_{HFC, t} \leq H_{HFC, \max} \\ \lambda_{l, HFC} \leq H_{HFC, t+1} - H_{HFC, t} \leq \lambda_{u, HFC} \end{cases} \quad (46)$$

$$K_{\min} \leq \frac{R_{HFC, t}}{P_{HFC, t}} \leq K_{\max}$$

where  $H_{HFC, \max}$ ,  $H_{HFC, \min}$  are the upper and lower limits of hydrogen power input into HFC,  $\lambda_{u, HFC}$ ,  $\lambda_{l, HFC}$  are the climbing upper and lower limits of HEC,  $K_{\max}$ ,  $K_{\min}$  are the upper and lower limits of the thermoelectric ratio of HFC.

#### 14) Hydrogen storage operation constraints

$$\begin{cases} 0 \leq H_{cha, t} \leq B_{cha, t} H_{\max} \\ 0 \leq H_{dis, t} \leq B_{dis, t} H_{\max} \\ H_{out, t} = \eta_{cha} H_{cha, t} - \eta_{dis} H_{dis, t} \\ S_{H, t} = S_{H, t-1} + H_{out, t} / S_{cap} \\ S_{H, 1} = S_{H, T} \\ B_{cha, t} + B_{dis, t} = 1 \\ S_{H, \min} \leq S_{H, t} \leq S_{H, \max} \end{cases} \quad (47)$$

where  $H_{cha, t}$  and  $H_{dis, t}$  are the flushing power of the hydrogen energy storage at time  $t$  respectively.  $B_{cha, t}$  and  $B_{dis, t}$  are the flushing state parameters of the hydrogen energy storage device at time  $t$ , respectively. Both are binary variables. When  $B_{cha, t}=1$  and  $B_{dis, t}=0$ , it represents the hydrogen storing state. When  $B_{cha, t}=0$  and  $B_{dis, t}=1$ , it represents the hydrogen releasing state.  $H_{out, t}$  is the output power of the hydrogen energy storage device at time  $t$ .  $\eta_{cha}$  and  $\eta_{dis}$  are the flushing efficiency of the hydrogen energy storage device at time  $t$  respectively.  $S_{H, t}$  is the capacity of the hydrogen energy storage device at time  $t$ .  $S_{cap}$  is the rated capacity of the hydrogen energy storage device.  $S_{H, \max}$  and  $S_{H, \min}$  are the upper and lower limits of the capacity of the hydrogen energy storage device, respectively.

## 5. Example analysis

Taking a comprehensive park as an example, an example analysis is carried out to verify the effectiveness of the proposed low-carbon economic dispatch model. The predicted of electricity, heat, gas, cold load as well as WT power, PV power are shown in Fig. 3. System parameters are shown in Table 2. In this paper, IES mathematical model is constructed by YALMIP in MATLAB and solved by IPOPT solver.

### 5.1. P2G two-stage operation process and CCS benefit analysis

In order to analyze and refine the two-stage operation process of P2G and the operation efficiency of CCS in IES, based on the ladder carbon trading mechanism, three cases are set up.

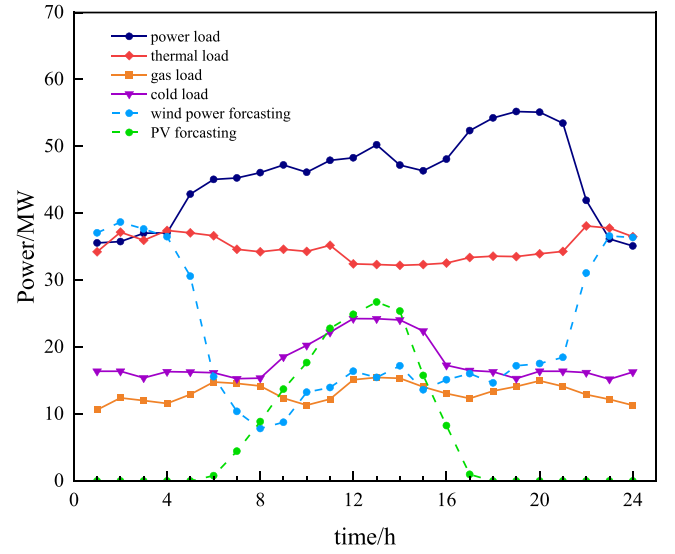


Fig. 3. Load demand and forecast output of wind and photovoltaics.

Table 2  
Scheduling results of each case.

| Parameter              | Value  | Parameter              | Value |
|------------------------|--------|------------------------|-------|
| $P_{P2G, \min}$ (MW)   | 0      | $P_{P2G, \max}$ (MW)   | 15    |
| $P_{CCS, \min}$ (MW)   | 0      | $P_{CCS, \max}$ (MW)   | 10    |
| $P_{CHP, \min}$ (MW)   | 10     | $P_{CHP, \max}$ (MW)   | 35    |
| $R_{CHP, \min}$ (MW)   | 0      | $R_{CHP, \max}$ (MW)   | 40    |
| $R_{CHP0}$ (MW)        | 5      | $P_{MT, \min}$ (MW)    | 5     |
| $P_{MT, \max}$ (MW)    | 30     | $P_{er, \min}$ (MW)    | 0     |
| $P_{er, \max}$ (MW)    | 4      | $Q_{s, \min}$ (MW)     | 0     |
| $Q_{s, \max}$ (MW)     | 30     | $H_{MR, \min}$ (MW)    | 0     |
| $H_{MR, \max}$ (MW)    | 25     | $H_{HFC, \min}$ (MW)   | 0     |
| $H_{HFC, \max}$ (MW)   | 25     | $S_{H, \min}$ (MW)     | 4     |
| $S_{H, \max}$ (MW)     | 18     | $b_f$                  | 25    |
| $\sigma$ (MWh/t)       | 0.269  | $\chi$ (t/MWh)         | 1.02  |
| $\eta_{H2}$            | 0.88   | $\eta_{MR}$            | 0.55  |
| $\eta_{HFC}$           | 0.9    | $\eta_{HFC}$           | 0.9   |
| $Z_{v1}$               | 0.15   | $Z_{v2}$               | 0.2   |
| $S_m$                  | 0.85   | $\varepsilon_1$ (t/MW) | 0.89  |
| $\varepsilon_2$ (t/MW) | 0.0017 | $\varepsilon_3$ (t/MW) | 26.15 |
| $\varepsilon_4$ (t/MW) | 1.09   | $\xi$ (¥/t)            | 150   |
| $g$                    | 0.39   | $\tau$                 | 0.25  |
| $m_1$ (¥/MW)           | 154    | $r_1$ (¥/MW)           | 350   |
| $m_2$ (¥/MW)           | 154    | $s_1$ (¥/MW)           | 210   |
| $a$ (¥/MW)             | 93.03  | $b$ (¥/MW)             | 0.028 |
| $c$ (¥/MW)             | 273    | $\eta_{MT}$            | 0.6   |
| $\eta_{hl}$            | 0.95   | $\eta_h$               | 1.9   |
| $\eta_w$               | 0.05   | $\eta_c$               | 2.4   |
| $c_1$ (¥/MW)           | 60     | $c_2$ (¥/MW)           | 120   |
| $c_3$ (¥/MW)           | 120    | $c_4$ (¥/MW)           | 26    |
| $c_5$ (¥/kg)           | 0.8    | $a_{r, \min}$          | -0.1  |
| $a_{r, \max}$          | 0.1    | $a_{c, \min}$          | -0.1  |
| $a_{c, \max}$          | 0.1    | $\lambda_l$            | -20   |
| $\lambda_u$            | 20     | $\lambda_{l, MT}$      | -20   |
| $\lambda_{u, MT}$      | 20     | $\lambda_{l, EL}$      | -4    |
| $\lambda_{u, EL}$      | 4      | $\lambda_{l, MR}$      | -5    |
| $\lambda_{u, MR}$      | 5      | $\lambda_{l, HFC}$     | -5    |
| $\lambda_{u, HFC}$     | 5      | $K_{\min}$             | 0.5   |
| $K_{\max}$             | 2.1    |                        |       |

**Case 1.** Consider CHP, regardless of CCS and P2G.

**Case 2.** Consider CHP, CCS and traditional P2G.

**Case 3.** Consider CHP, CCS and refine the P2G of two-stage operation process.

The scheduling results of the above three cases are shown in Table 3.

**Table 3**  
Scheduling results of each case.

|                               | Case 1   | Case 2   | Case 3   |
|-------------------------------|----------|----------|----------|
| Carbon emissions/kg           | 1360.92  | 1238.71  | 1005.27  |
| Carbon trading cost/¥         | 8692.67  | 5026.46  | 3.87     |
| Wind energy utilization rate  | 64.68 %  | 97.24 %  | 96.08 %  |
| Photovoltaic utilization rate | 61.96 %  | 100 %    | 99.75 %  |
| Total cost/¥                  | 39583.09 | 34916.88 | 26317.04 |

#### 5.1.1. P2G two-stage operation process benefit analysis

The advantages of the traditional P2G model refined into an electrolytic cell, methane converter, hydrogen fuel cell and hydrogen storage device are analyzed through **Case 1**, **Case 2** and **Case 3**. It can be seen from **Table 3** that the total operating cost and carbon emissions of **Case 1** are the highest, which are ¥39583.09 and 1360.92 kg respectively, and the WT power and PV power consumption capacity are the weakest, which are 64.68 % and 61.96 % respectively. **Case 2** introduces P2G technology, the total operating cost of the system compared with **Case 1** decreased by ¥34916.88, carbon emissions decreased by 122.21 kg, the accommodation capacity of WT and PV power increased by 32.56 % and 38.04 % respectively. In **Case 3**, the P2G two-stage process is refined, and the WT and PV power consumption capacity are reduced to 96.08 % and 99.75 % respectively. Although it is reduced by 1.16 % and 0.25 % compared with **Case 2**, the total operating cost of the system is further reduced to ¥26317.04. And carbon emissions decreased by 18.85 % compared with **Case 2**.

Based on the above analysis, it can be concluded that the P2G model refined into a combination of electrolytic cell, methane converter, hydrogen fuel cell and hydrogen storage device can realize the efficient utilization of hydrogen energy and reduces the combustion of natural gas, thus reducing the operating cost and carbon emissions of the system.

#### 5.1.2. CCS benefit analysis

It can be seen from **Table 3** that the total cost of **Case 2** decreased by ¥4666.21 Compared with **Case 1**. And the Carbon emissions cost is decreased by ¥3666.21. Hence, the introduction of P2G and CCS in IES is conducive to the low-carbon economic operation of the system. The reason is that CCS captures CO<sub>2</sub> produced by CHP and micro gas turbines and provides it as a carbon source to P2G, thereby reducing the carbon emissions and operating costs of the system.

#### 5.1.3. Analysis of new energy consumption capacity of each case

The WT power consumption of the three cases in each period is shown in **Fig. 3(a)**. In the periods of 0.00–5.00 and 22.00–24.00, the wind resources are abundant and the WT power abandonment phenomenon is frequent. Compared with **Case 1**, **Case 2** introduces P2G and

CCS. CCS provides a carbon source for P2G, promotes P2G to convert surplus WT power into electrical energy, improves WT power consumption rate. Compared with **Case 2**, although the WT power consumption rate of **Case 3** decreases slightly, the overall economic and environmental benefits of the system are greatly improved, which does not affect the overall operation of the system.

The PV power consumption of each case in each period is shown in **Fig. 3(b)**. Compared with **Case 1**, **Case 2** has enhanced PV power consumption capacity. Compared with **Case 2**, although the PV power consumption rate in **Case 3** is slightly reduced, the overall economic and environmental benefits of the system are greatly improved, which does not affect the overall operation of the system (**Fig. 4**).

#### 5.1.4. Operational result analysis of case 4

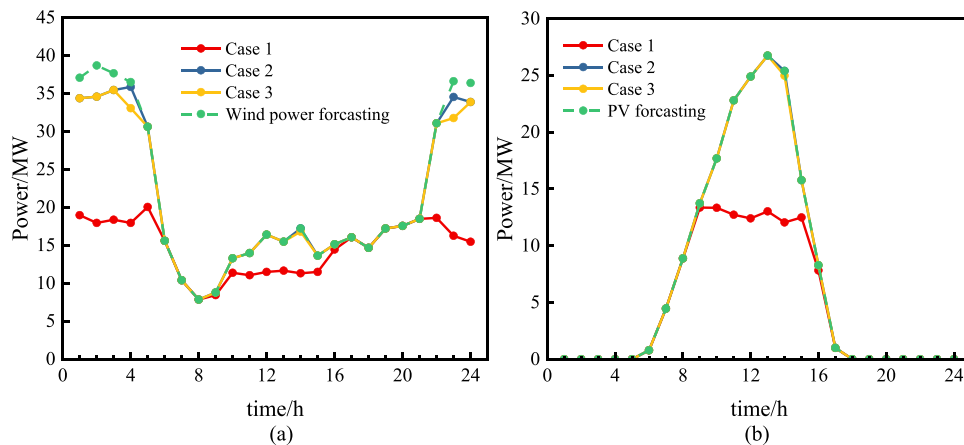
In order to verify the feasibility and advantages of the model proposed in this paper, the simulation results of **Case 3** are further analyzed. The IES power operation results of **Case 3** are shown in **Fig. 5**.

It can be seen from **Fig. 5(a)** that the electrical load and the electrical power required by ER are provided by CHP, MT, WT, PV and HFC. 0.00–5.00, 10.00–15.00 and 22.00–24.00 are the peak periods of renewable energy, and the WT and PV provide most of the electrical power. P2G converts the surplus power to weaken the electro-thermal coupling characteristics of CHP. In the low peak period of renewable energy, electrical load demand increases, in order to ensure the stability of power supply, CHP, HFC power output increases.

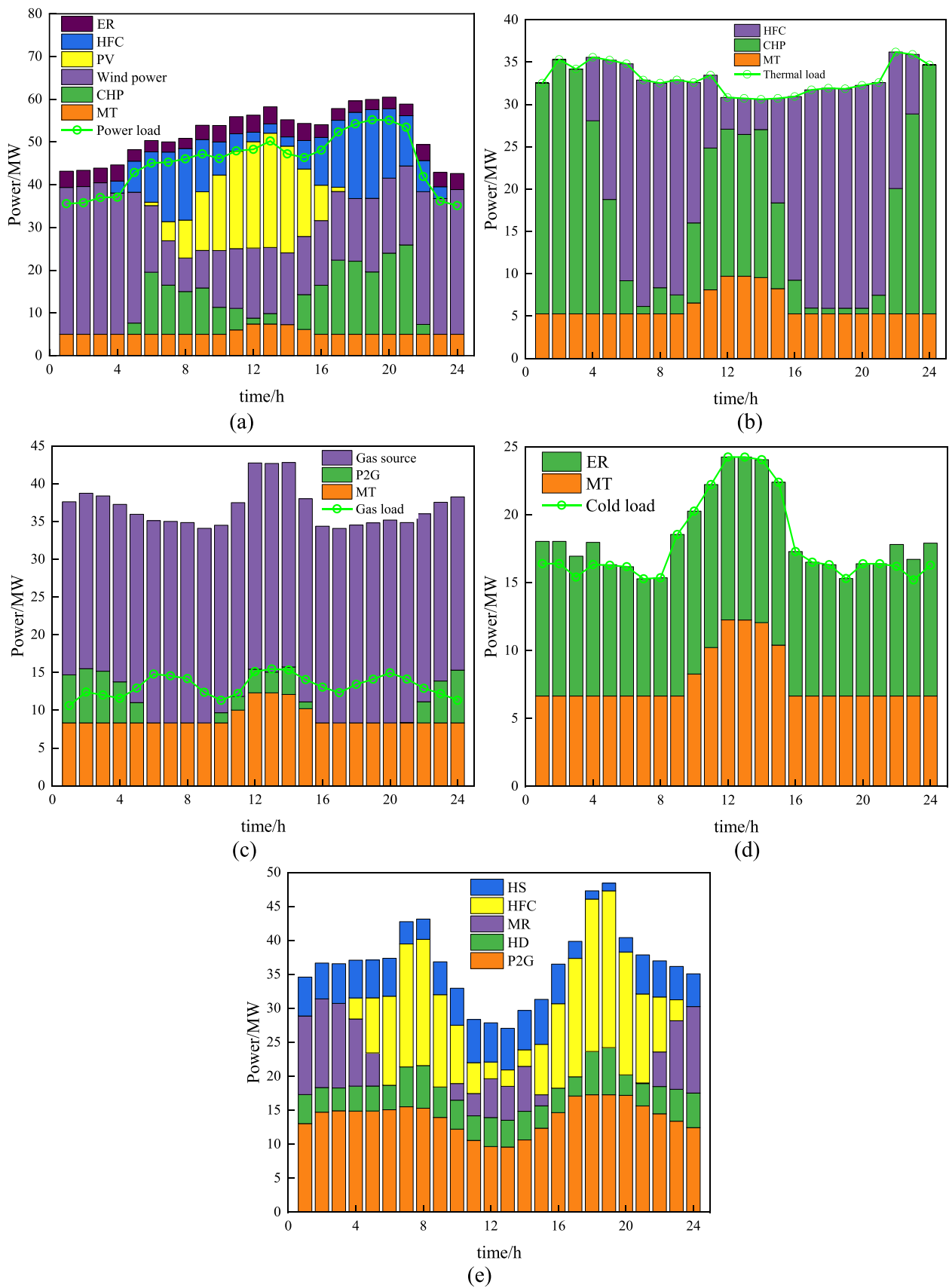
**Fig. 5(b)** shows that the heating power for balancing the heating load is provided by MT, CHP and HFC. In the period of high renewable energy, CHP outputs thermal power. In the period of low renewable energy, because WT and PV are difficult to meet the demand of electrical load, CHP outputs electric power to ensure the reliability of the system power supply. At this time, the heating power is mainly provided by HFC.

It can be seen from **Fig. 5(c)** that during the peak period of renewable energy, P2G converts the electrical power into the gas power to provide a gas source for MT and gas load. At this time, the output power of the natural gas source decreases, the operating cost decreases, and the economy of system operation is improved.

It can be seen from **Fig. 5(e)** that in the period of high renewable energy, the hydrogen produced by EL is converted into natural gas by MR to supply gas load, or stored in hydrogen storage equipment. In the period of low renewable energy, hydrogen storage equipment releases hydrogen energy, and HFC directly converts hydrogen energy into the electrical power or heating power to achieve efficient utilization of hydrogen energy.



**Fig. 4.** Power accommodation analysis of different scenarios for WT and PV. (a) WT power accommodation of each case; (b) PV power accommodation of each case.



**Fig. 5.** IES power scheduling result of Case 4. (a) Electrical power; (b) Thermal power; (c) Gas power; (d) Cold power; (e) Hydrogen power.



5.2. Benefit analysis of ladder carbon trading mechanism

In order to verify the superiority of the stepped carbon trading mechanism, based on the IES low-carbon economic dispatch model with CCS-P2G-CHP, two operating scenarios are set up for comparative analysis.

**Scenario 1.** : IES scheduling model without carbon trading mechanism;

**Scenario 2.** : IES scheduling model with traditional carbon trading mechanism;

**Scenario 3.** : IES scheduling model with ladder carbon trading mechanism.

The results for both scenarios are shown in Table 4.

As can be seen from Table 4, the carbon emissions of Scenario 3 are 1137.01 kg lower than that of Scenario 1 and Scenario 2, and the carbon transaction cost is ¥3.87.

Since Scenario 1 does not consider the carbon trading mechanism, in the optimal scheduling process of IES, only the total energy consumption cost other than carbon trading cost is taken as the optimization objective. Concurrently, the system burns a large amount of natural gas with low price for energy supply, resulting in high carbon emissions of CHP units and MT. Therefore, Scenario 1 has the highest carbon emissions and the largest carbon trading cost. Scenario 2 considers the traditional carbon trading mechanism, which considers the impact of carbon trading costs on the optimal scheduling of IES, and improves the low-carbon operation of the system to a certain extent. Compared with Scenario 1, the carbon emissions are reduced by 16.98 %, and the carbon trading costs are reduced to 2738.12. However, the traditional carbon trading mechanism adopts a fixed carbon price, which limits the role of carbon emission reduction. Scenario 3 employs the ladder carbon trading mechanism, both carbon emissions and carbon trading costs are lower compared with Scenario 2. This derives from that with the increase of carbon emissions, the carbon price increases step by step, which causes the system to adjust the output of each unit to achieve low-carbon operation, and the benefits of selling surplus quotas are higher than the increased costs of using the ladder carbon price.

6. Conclusions

This paper establishes an IES low-carbon economic dispatch model with CCS-P2G-CHP coupling. On the basis of CCS-P2G-CHP coupling, the traditional P2G is refined into two processes of electricity to hydrogen production and methanation. Moreover, the coupling of CCS-P2G-CHP can not only reduce the operation constraint of CHP but also realize the recycling of carbon and reduce the energy cost of the system. Then a ladder carbon trading mechanism model is introduced to alleviate the contradiction between the rapid development of Renewable energy and the difficulty of consumption, and improve the economic and environmental benefits of the system. Example analyses verify the effectiveness of the proposed dispatch model on the low-carbon development of the system. With the development of energy storage technologies and improvement of related market mechanisms, it is worth to study the benefits of applying energy storage technologies in the proposed model and the impacts of the regulatory changes on the operation of IES in the future.

CRediT authorship contribution statement

**Chen Yang:** Conceptualization, Writing – original draft. **Xiaofeng Dong:** Validation, Writing – review & editing. **Gang Wang:** Validation, Writing – review & editing. **Dianrui Lv:** Conceptualization, Validation. **Ren Gu:** Writing – review & editing. **Yuanqing Lei:** Writing – review & editing.

**Table 4**  
Scheduling results of each Scenario.

|                       | Scenario 1 | Scenario 2 | Scenario 3 |
|-----------------------|------------|------------|------------|
| Carbon emissions/kg   | 1469.27    | 1137.01    | 1005.27    |
| Carbon trading cost/¥ | 3551.44    | 2738.12    | 3.87       |

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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